# Windows Fundamentals for SOC Analysts

Detailed study report focused only on the Windows concepts required to build a strong SOC foundation

**Scope:** Windows architecture, processes & services, users & permissions, file system, registry, PowerShell, and Windows Security. The report deliberately excludes unrelated Windows administration and end-user topics.

**How to use this report:** Study each section, reproduce the hands-on commands in a Windows VM, and connect every concept to the question: *"What evidence would a SOC analyst see if an attacker abused this component?"*

## 1. What You Actually Need to Learn

| Area | SOC Objective | Priority |
|---|---|---|
| Windows architecture | Understand how major Windows components interact and where security telemetry originates. | High |
| Processes & services | Identify normal vs suspicious execution, parent-child relationships, persistence and service abuse. | Very High |
| Users & permissions | Understand identities, groups, privileges, UAC and access control. | Very High |
| File system | Understand NTFS, permissions, paths, metadata and common locations abused by attackers. | High |
| Registry | Understand configuration, persistence, execution and investigation locations. | Very High |
| PowerShell | Understand command execution, scripting, logging and common abuse patterns. | Very High |
| Windows Security | Understand authentication, Defender, firewall, security controls and security telemetry. | Very High |

## 2. Windows Architecture

**Goal:** You do not need to become a Windows kernel developer. You need enough architecture knowledge to understand processes, privileges, system calls, security boundaries and where endpoint activity comes from.

### 2.1 User Mode vs Kernel Mode

- **User mode:** Applications and many Windows services operate with restricted privileges. Examples include browsers, Office applications and many user processes.
- **Kernel mode:** Core operating-system components, kernel code and drivers operate with much higher privilege.
- **SOC relevance:** Malware attempting to gain higher privileges, load drivers or interact with protected components can create important detection signals.

### 2.2 Major Windows Components

- **Windows Kernel:** Provides core operating-system functionality.
- **Executive:** Coordinates components such as process/thread management, memory, I/O and security-related functions.
- **Services:** Background components that can run without an interactive user session.
- **Winlogon / authentication components:** Participate in user logon and session initialization.
- **LSASS:** A security-sensitive Windows process involved in authentication and credential-related operations. Unexpected access to it is a major investigation signal.
- **Windows Defender / security components:** Provide endpoint protection and telemetry.

### 2.3 Processes, Threads and Handles

- **Process:** A running program with its own virtual address space and security context.

- **Thread:** A unit of execution inside a process.
- **Handle:** A reference used by a process to access operating-system objects such as files, registry keys and processes.

SOC question to ask: **Which process started this process, under which account, with what command line, and what did it access?**

# 3. Processes & Services

## 3.1 Processes — Concepts to Know

- Process name and executable path
- PID (Process ID)
- Parent process / child process relationship
- User account running the process
- Command-line arguments
- Integrity level and privileges
- Network connections associated with a process
- Loaded modules/DLLs at a conceptual level
- Process creation and termination

## 3.2 Normal Windows Processes You Should Recognize

| Process | Why you should recognize it |
|---|---|
| explorer.exe | Windows shell / desktop environment. |
| svchost.exe | Hosts Windows services; many legitimate instances exist. |
| services.exe | Service Control Manager process. |
| lsass.exe | Security authority process; highly security-sensitive. |
| winlogon.exe | Windows logon/session component. |
| csrss.exe | Client/Server Runtime Subsystem; critical Windows process. |
| smss.exe | Session Manager; early Windows initialization. |
| System | Kernel/system activity; investigate carefully rather than assuming maliciousness. |
| powershell.exe | PowerShell process; legitimate but frequently abused. |
| cmd.exe | Command Prompt; legitimate but can execute malicious commands. |

## 3.3 Services

- Understand what a Windows service is and why services can start automatically.
- Know the Service Control Manager (SCM) concept.
- Understand service name vs display name.
- Know that attackers may abuse services for persistence or execution.
- Learn to investigate unusual service creation, executable paths, accounts and start types.

```
Get-Service
Get-Service | Where-Object {$_.Status -eq 'Running'}
Get-CimInstance Win32_Service | Select Name,State,StartMode,StartName,PathName
```

## 3.4 SOC Investigation Checklist — Process

- Is the executable located in an expected directory?
- Is the filename pretending to be a legitimate process?
- What is the parent process?
- What account executed it?
- What command line was used?
- Did it create network connections?
- Did it create files, scheduled tasks, services or registry changes?

# 4. Users, Groups & Permissions

## 4.1 Identity Basics

- Local user accounts vs domain accounts
- Security groups and group membership
- Administrator vs standard-user context
- Service accounts and why their privileges matter
- Security identifiers (SIDs) at a conceptual level

## 4.2 Access Control

- Windows uses access control lists (ACLs) to control access to securable objects.
- **ACE:** An access-control entry describing an allow/deny permission for a security principal.
- Know common permissions: read, write, modify, execute and full control.
- Understand inheritance conceptually: permissions can flow from parent objects to children.

## 4.3 Privileges and UAC

- **UAC:** User Account Control helps reduce the risk of unauthorized elevation of privileges.
- Understand the difference between being an administrator and running a process with elevated privileges.
- Know the idea of privilege escalation: obtaining permissions beyond the attacker's initial access.

## 4.4 Commands for Practice

```
whoami
whoami /all
whoami /groups
whoami /priv
whoami /user
net user
net localgroup administrators
Get-LocalUser
Get-LocalGroup
Get-LocalGroupMember Administrators
```

SOC questions: **Who performed the action? What account was used? Was the account privileged? Was privilege elevation involved?**

# 5. Windows File System

## 5.1 NTFS Fundamentals

- Understand files, directories, paths and file extensions.
- Understand NTFS permissions and inheritance.

- Know that file metadata and timestamps can support investigations.
- Understand that a file's extension does not prove what the file actually is.

## 5.2 Important Locations for SOC Analysts

| Location | SOC relevance |
|---|---|
| C:\Windows\System32 | Core Windows binaries; unexpected modifications or look-alikes deserve investigation. |
| C:\Windows\Temp | Temporary files; can contain artifacts from installers or malicious activity. |
| %TEMP% | User temporary directory; commonly used by legitimate software and malware. |
| %APPDATA% | User application data; commonly used by applications and sometimes abused for persistence. |
| %LOCALAPPDATA% | Per-user local application data. |
| Downloads | Common location for downloaded files and initial user-delivered payloads. |
| Startup folders | Potential persistence location. |

## 5.3 File Investigation Commands

```
Get-ChildItem C:\Windows\Temp
Get-Item 'C:\path\file.exe' | Format-List *
Get-FileHash 'C:\path\file.exe' -Algorithm SHA256
Get-Acl 'C:\path\file.exe' | Format-List
```

- Learn to identify file path, size, timestamps, hash, owner and ACLs.
- For suspicious files, compare path + signer + hash + process context rather than judging by filename alone.

# 6. Windows Registry

## 6.1 What the Registry Is

- A hierarchical configuration database used by Windows and applications.
- It contains settings for the operating system, users, applications and security-related configuration.
- Registry data is organized into hives, keys, subkeys and values.

## 6.2 Main Hives to Recognize

- **HKLM (HKEY_LOCAL_MACHINE):** Machine-wide configuration.
- **HKCU (HKEY_CURRENT_USER):** Configuration for the current user.
- **HKCR:** Combined view related to classes/file associations.
- **HKU:** User profile information for loaded user accounts.

## 6.3 Registry and Persistence

- Learn the concept of Run/RunOnce persistence locations.
- Understand that attackers may modify registry configuration to execute code automatically.
- Do not memorize every registry key; learn the major persistence categories and how to investigate changes.

```
reg query HKCU\Software\Microsoft\Windows\CurrentVersion\Run
reg query HKLM\Software\Microsoft\Windows\CurrentVersion\Run
Get-ItemProperty 'HKCU:\Software\Microsoft\Windows\CurrentVersion\Run'
```

SOC investigation: identify **what value changed, what executable/script it points to, who changed it, when it changed, and whether the target is legitimate.**

# 7. PowerShell

## 7.1 Why PowerShell Is Critical for SOC

- PowerShell is a legitimate Windows automation and administration framework.
- Because it is powerful and widely available, attackers may use it for execution, discovery, downloading, persistence and lateral movement.
- The SOC objective is not to block PowerShell blindly; it is to identify suspicious usage and context.

## 7.2 PowerShell Fundamentals

- Cmdlets and verbs/nouns
- Objects and pipelines
- Variables
- Scripts (.ps1)
- Execution context and privileges
- Modules
- Encoded commands as an investigation clue
- Remote execution concepts

## 7.3 Commands You Should Be Comfortable Reading

```
Get-Process
Get-Service
Get-ChildItem
Get-ItemProperty
Get-ComputerInfo
Get-NetIPConfiguration
Get-NetTCPConnection
Get-WinEvent -LogName Security -MaxEvents 20
```

## 7.4 Logging Concepts

- PowerShell operational logging
- Script Block Logging
- Module Logging
- Transcription
- Understand that logging configuration affects the evidence available to a SOC analyst.

## 7.5 Suspicious PowerShell Indicators

- Unexpected PowerShell launched by an unusual parent process.
- Highly obfuscated or encoded commands.
- Commands that download or execute remote content.
- PowerShell running under an unexpected account or from an unusual application.
- Activity that combines PowerShell with persistence, credential access or lateral movement.

Important: a suspicious-looking PowerShell command is an investigation lead, not automatically proof of malicious activity.

# 8. Windows Security

## 8.1 Authentication

- Understand authentication vs authorization.
- Know the basic purpose of NTLM and Kerberos.
- Understand logon sessions and authentication events at a conceptual level.
- Learn why repeated failed logons, unusual successful logons and abnormal authentication patterns matter to a SOC.

## 8.2 Windows Security Architecture

- Security accounts and security principals
- Access tokens
- Privileges
- UAC
- Security policy
- Protected processes at a conceptual level

## 8.3 Windows Defender

- Understand antivirus vs broader endpoint detection and response concepts.
- Learn what detections, alerts, quarantine and remediation mean.
- Understand that Defender telemetry can contribute to an investigation alongside Windows events.

## 8.4 Windows Firewall

- Understand inbound vs outbound rules.
- Know profiles conceptually: Domain, Private and Public.
- Understand how firewall activity can support network investigations.

```
Get-NetFirewallProfile
Get-NetFirewallRule | Select DisplayName,Enabled,Direction,Action
```

## 8.5 Security Event Logs

| Event ID | Meaning / SOC relevance |
|---|---|
| 4624 | Successful logon. |
| 4625 | Failed logon. |
| 4672 | Special privileges assigned to a new logon. |
| 4688 | A new process has been created. |
| 4698 | A scheduled task was created. |
| 4720 | A user account was created. |
| 4728 / 4732 | User added to a privileged/security group (depending on group type). |
| 4740 | Account was locked out. |
| 4768 | Kerberos authentication ticket requested. |
| 4769 | Kerberos service ticket requested. |
| 7045 | A new service was installed. |

# 9. Hands-On Windows Lab

Use a Windows virtual machine for practice. Do not perform security testing on systems you do not own or have explicit permission to test.

## Lab 1 — Process Investigation

- Open Task Manager and identify running processes.
- Use PowerShell to list processes and sort by CPU or memory.
- Pick a legitimate process and inspect its executable path and account.
- Use Sysinternals Process Explorer later to compare what additional process information is available.

```
Get-Process | Sort-Object CPU -Descending | Select-Object -First 10
```

## Lab 2 — Users and Privileges

```
whoami /all
Get-LocalUser
Get-LocalGroupMember Administrators
```

- Identify your current user, groups and privileges.
- Understand which account is privileged and why.

## Lab 3 — File Investigation

```
Get-ChildItem $env:TEMP
Get-FileHash "$env:WINDIR\System32\notepad.exe" -Algorithm SHA256
Get-Acl "$env:WINDIR\System32\notepad.exe"
```

## Lab 4 — Registry

```
reg query HKCU\Software\Microsoft\Windows\CurrentVersion\Run
```

- Document what startup entries exist on your lab machine.

## Lab 5 — Event Viewer

- Open Event Viewer.
- Navigate to Windows Logs → Security.
- Find a successful logon (4624) and failed logon (4625) if available.
- Record timestamp, account, logon type and source information where present.

## Lab 6 — PowerShell Telemetry

- Run harmless PowerShell commands and observe what appears in PowerShell logs when logging is enabled.
- Learn to distinguish command content, process information and authentication context.

# 10. SOC Investigation Thinking

For every Windows alert, train yourself to answer these questions:

- Who — which account or security principal performed the action?
- What — what process, command, file, service or registry value was involved?
- Where — on which host and from which path/source did it occur?
- When — exact timestamp and sequence of events?
- How — parent process, command line, authentication method or execution mechanism?

- Why — is there a legitimate business explanation?
- What next — what additional telemetry should be checked?

# 11. Recommended Study Sequence

| Week/Stage | Study | Outcome |
|---|---|---|
| 1 | Windows architecture + processes | Understand execution and process relationships. |
| 2 | Services + users + groups + permissions | Understand identity, privilege and persistence. |
| 3 | File system + important paths | Investigate suspicious files and locations. |
| 4 | Registry | Understand configuration and common persistence mechanisms. |
| 5 | PowerShell fundamentals | Read and investigate PowerShell activity. |
| 6 | Windows Security + authentication | Understand security boundaries and authentication. |
| 7 | Event Logs + important IDs | Read endpoint telemetry. |
| 8 | Hands-on investigation | Correlate processes, users, files, registry and logs. |

# 12. What to Exclude for Now

To keep this report aligned with your SOC goal, these areas are intentionally **not** treated as core study requirements:

- Deep Windows kernel-driver development
- Windows application development
- Advanced Active Directory administration
- Group Policy administration in depth
- Windows deployment/imaging
- Enterprise backup and storage administration
- Desktop customization and end-user troubleshooting
- Deep Windows networking configuration
- Kernel debugging and reverse engineering

These can become useful later, especially Active Directory, Sysmon, EDR and advanced Windows internals, but they are outside the requested Windows Fundamentals scope.

# 13. After Completing This Report

Move next into **Sysmon** → **Microsoft Defender** → **Defender for Endpoint/XDR** → **KQL** → **SOC investigation labs**. At that point, Windows concepts should be learned through real alerts and telemetry rather than only theory.

Core reference: Microsoft Learn — Windows: https://learn.microsoft.com/en-us/windows/

Microsoft Sysinternals: https://learn.microsoft.com/en-us/sysinternals/

Microsoft Defender XDR: https://learn.microsoft.com/en-us/defender-xdr/